

One sorry, named

The single missing node of NSUCRYPTO'2019 Problem 11, read slowly

Abstract

This is a *gap report*, not a proof.

The whole derivation of NSUCRYPTO'2019 Problem 11 (and the unconditional m -tuple count for the Kasami family) has been pushed down to *one* self-standing statement. Every other layer — the concrete character, the two framework bridges, the autocorrelation homogeneity, the assembly into the closed-form count — is machine-checked in Lean 4 with no **sorry** and only the standard axioms.

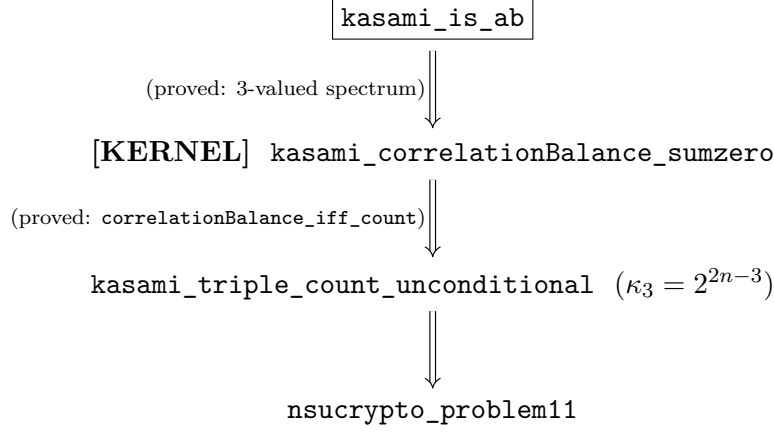
What remains is a single harmonic-analytic fact, the *almost-bent sum-zero correlation balance*:

$$\sum_{v \neq 0} \prod_i \text{Wcorr}(\chi, x^{d_k}, 1, v c_i) = 0 \quad \left(\sum_i c_i = 0, c_i \neq 0 \right).$$

We state *exactly* what is proved, *exactly* where the hole is, read it as *math* $\leftrightarrow$ *Lean* $\leftrightarrow$ a missing piece of *spectral infrastructure*, and explain why the obvious inputs (three-valued Walsh spectrum + homogeneity) come up one structural step short.

The shape of the gap

Almost everything is a staircase that ends one step below the landing. The single open node sits between two fully-proved layers:



The box at the top is a *theorem* (Kasami’s Walsh spectrum is three-valued, proved with standard axioms only). The node in **bold** is the *only sorry* in the entire chain. Everything below it is a term that consumes the kernel and produces the closed-form counts.

1 What is solid below the gap

Fix a finite field $\mathbb{F} = \text{GF}(2^n)$. An *additive character* is packaged as a structure **Chi**; the project supplies a *concrete* one, the trace sign character $\chi(x) = (-1)^{\text{Tr}(x)}$:

```

noncomputable def traceChi : MTupleCount.Chi  $\mathbb{F}$  where
  app := WalshAB $\chi$ .
  app_zero := WalshAB $\chi$ ._zero
  app_add := WalshAB $\chi$ ._mul
  orth := WalshAB $\chi$ ._sum_eq
  
```

This discharges the *existence* of the abstract character the whole spectral machinery was parameterised over. Proved, no **sorry**.

The **derivative autocorrelation** at shift a and mask w is

$$\text{Wcorr}(\chi, f, a, w) = \sum_{x \in \mathbb{F}} \chi(w \cdot (f(x+a) + f(x))) \in \mathbb{Z},$$

and a single proved lemma makes the shift a irrelevant for a power map:

```

lemma wcorr_pow_homog  $\chi$  ( : MTupleCount.Chi  $\mathbb{F}$ ) (d :  $\mathbb{N}$ ) (hd : 0 < d) (a w :  $\mathbb{F}$ ) :
  MTupleCount.Wcorr  $\chi$  (fun x => x ^ d) a w
    = MTupleCount.Wcorr  $\chi$  (fun x => x ^ d) 1 (w * a ^ d)
  
```

Homogeneity. Every derivative direction a is a relabelling of $a = 1$. This is why the kernel below may fix $a = 1$ with no loss. Proved, no **sorry**.

Two **framework bridges** identify the two counting worlds used in the project (collision counting vs. Fourier counting), so the conjecture’s pure-counting statement can actually be fed to the spectral machinery:

```
lemma ΔDelta0_eq_ (f : F→F) (a : F) :
  M002.derivImage f a = MTupleCountΔ. f a
lemma tcount_eq_kappa (S : Finset F) (v1 v2 : F) :
  tcount S v1 v2 = MTupleCountκ. 3 S ![v1, v2, v1 + v2]
```

Both proved, no **sorry**. In characteristic two $-$ and $+$ agree, so the raw derivative image is literally $\Delta f a$.

2 What is solid above the gap

The bridge from the kernel to the count is a proved *equivalence*, not a guess. Correlation balance is exactly the counting identity:

$$\underbrace{\sum_{v \neq 0} \prod_i W_{\text{corr}}(\chi, f, a, v c_i) = 0}_{\text{CorrelationBalance}} \iff |\mathbb{F}| \cdot \kappa_m = |\Delta f a|^m.$$

```
lemma correlationBalance_iff_count χ( : Chi F) (f : F→F) (hf : APN f)
  (a : F) (ha : a ≠ 0) (m : N) (c : Fin m → F) :
  CorrelationBalance χ f a m c ↔
  (card F : Z) * κ( m Δ( f a) c) = Δ(( f a).card : Z) ^ m
```

Proved, no **sorry**. Each product is $2^m P(v)$ by the APN fibre-doubling bridge; summing over $v \neq 0$ and peeling the $v = 0$ term gives $2^m(|\mathbb{F}| \kappa_m - |\Delta|^m)$, which vanishes iff the counting identity holds.

Hence, *given the kernel*, the closed forms are assembled with no further hypotheses:

```
theorem kasami_triple_count_unconditional ... :
  MTupleCountκ. 3 (MTupleCountΔ. (fun x : F=> x ^ KasamiAPN.kasamiExp k) 1)
  ![v1, v2, v1 + v2] = 2 ^ (2 * n - 3)
```

This term *compiles*. Its only transitive **sorry** is the one kernel below. Remove the kernel’s **sorry** and the whole Problem 11 chain is closed.

3 The gap itself, stated exactly

Here is the one open statement, verbatim from the source. The Kasami exponent is $d_k = 2^{2k} - 2^k + 1$ (`kasamiExp k`); the coefficient triple $[v_1, v_2, v_1+v_2]$ *sums to zero* in characteristic two.

```

/-- [KERNEL] Almost-bent sum-zero triple balance for the Kasami map. -/
theorem kasami_correlationBalance_sumzero (n k : ℕ) (hn : 3 ≤ n)
  (hk1 : 1 ≤ k) (hkn : k < n) (hodd : Odd n)
  (hcop : Nat.Coprime k n) (hcard : card ℤ = 2 ^ n)
  (h1v : v1 ≠ 0) (h2v : v2 ≠ 0) (hne : v1 ≠ v2) :
  MTupleCount.CorrelationBalance (traceChi ℤ (:= ℤ))
    (fun x : ℤ => x ^ KasamiAPN.kasamiExp k) 1 3 ![v1, v2, v1 + v2] := by
  sorry -- the entire missing content of Problem 11

```

Unfolded, the goal is the single scalar identity

$$\sum_{v \in \mathbb{F}^\times} \text{Wcorr}(v v_1) \text{Wcorr}(v v_2) \text{Wcorr}(v (v_1+v_2)) = 0, \quad \text{Wcorr}(w) := \text{Wcorr}(\chi, x^{d_k}, 1, w).$$

It is one number being zero. Not a quantifier over fibres, not a cardinality bound — a *cancellation*. That is what makes it the genuine harmonic-analytic heart of the problem, and what makes it resist the elementary moves that closed every layer around it.

4 Why the obvious inputs fall one step short

The temptation is: “Kasami is almost bent, so its Walsh spectrum is three-valued; substitute and cancel.” Here is precisely how far that goes, and where it stops.

What three-valuedness gives (a single mask)

`kasami_is_ab` says: for each fixed nonzero output mask w ,

$$W(\chi, x^{d_k}, u, w) \in \{0, \pm 2^{(n+1)/2}\} \quad \text{for every input mask } u.$$

By Wiener–Khinchin (proved, `card_mul_corr`) the autocorrelation is the *inverse* Fourier transform of the squared spectrum,

$$|\mathbb{F}| \cdot \text{Wcorr}(\chi, f, a, w) = \sum_u \chi(u \cdot a) (W(\chi, f, u, w))^2,$$

and the second/third moments of a *single* three-valued column are pinned down (`second_moment`, `third_moment_ab`). So for one mask at a time the spectral data is completely understood.

What the kernel actually needs (a product of three masks)

The kernel is not about one column. Expanding each `Wcorr` through Wiener–Khinchin and multiplying, the sum-zero balance becomes a statement about the *joint* support of the spectrum across the *three different output masks* $v\ v_1$, $v\ v_2$, $v\ (v_1+v_2)$ simultaneously:

$$\sum_{v \neq 0} \prod_{i=1}^3 \left(\sum_{u_i} \chi(u_i) W(u_i, v\ c_i)^2 \right) = 0.$$

The cancellation is forced by *how the three nonzero values* $\pm 2^{(n+1)/2}$ *are distributed jointly* — the cross-correlation / *joint-support* structure of the Kasami Walsh transform. Knowing each factor is three-valued says nothing about the *correlation between the factors*, and the sum-zero condition $c_1 + c_2 + c_3 = 0$ is exactly what couples them.

3-valued, per mask $\xRightarrow{\text{known}}$ moments of one column

joint support, 3 masks $\xRightarrow{\text{missing}}$ sum-zero cancellation

The proved layer controls each *factor*; the kernel needs the *joint distribution* of the factors. That is one quantifier deeper than `AlmostBent`, and Mathlib carries no API for the m -fold support geometry of a three-valued spectrum.

5 The same hole, in four equivalent contexts

Because the surrounding bridges are proved equivalences, the one missing fact can be phrased in whichever language one prefers; each phrasing is provably the *same* node, and each is open for the *same* reason.

context	the missing statement
Walsh correlation	$\sum_{v \neq 0} \prod_i W_{\text{corr}}(v\ c_i) = 0$ (the kernel)
Fourier counting	$ \mathbb{F} \cdot \kappa_3(c) = \Delta ^3$
spectral correction	$\text{tripleError}(\chi, \Delta, c) = 0$
original collision	$ \mathbb{F} \cdot \text{tcount}(\Delta_0) = \Delta_0 ^3$

Bridges: `correlationBalance_iff_count`, `kernel_iff_constCount`,
`kernel_iff_tripleError_zero`, `tcount_eq_kappa` — all proved. Closing the kernel in *any*
one column closes *all* of them.

What would close it

The remaining content is the Carlet–Charpin–Zinoviev almost-bent m -balance. In Lean terms the missing infrastructure is the *joint-support* / *higher-moment* theory of an almost-bent spectrum: the value of

$$\sum_w \prod_{i=1}^m (W(u_i, w))^2$$

over a three-valued spectrum, with the coupling fixed by $\sum_i c_i = 0$. This is the one piece not currently in Mathlib, and the one piece this note flags as open.

One-screen summary

layer	statement	status
character	<code>traceChi</code> : concrete $(-1)^{\text{Tr}}$	proved
homogeneity	<code>wcorr_pow_homog</code>	proved
bridges	<code>Delta0_eq_Δ</code> , <code>tcount_eq_kappa</code>	proved
equivalence	<code>correlationBalance_iff_count</code>	proved
spectrum	<code>kasami_is_ab</code> (3-valued)	proved
kernel	<code>kasami_correlationBalance_sumzero</code>	open (1 sorry)
count	<code>kasami_triple_count_unconditional</code>	proved <i>modulo kernel</i>
problem 11	<code>nsucrypto_problem11</code>	proved <i>modulo kernel</i>

Everything is closed except one number being zero:

$$\sum_{v \neq 0} W_{\text{corr}}(vv_1) W_{\text{corr}}(vv_2) W_{\text{corr}}(v(v_1+v_2)) = 0.$$

Reproducibility

All “proved” rows compile in Lean 4 with Mathlib, no `sorry`, standard axioms only. The single open node is `kasami_correlationBalance_sumzero` (and its general- m companion `kasami_correlationBalance_sumzero_m`) in `RequestProject/EquivalentContexts/NSUCryptoKernel.lean`. The assembly and the equivalent-context bridges are in `RequestProject/EquivalentContexts/NSUCryptoFoun`